

Arithmetic of the Herglotz function: verified tables, new rational-value theorems, and bridges to the golden ladders and the Clausen lattice

Vladimir Reshetnikov

2026-06-25

Consolidated research report. Created (UTC) 2026-06-25T21:43:04Z; repository **HEAD**
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Source paper: D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*
(src/PolyLog/docs/pdfs/HerglotzFunction/).

Stores: herglotz-rational-values.wl, herglotz-quadratic-units.wl. Reproducers:
src/PolyLog/tools/scratch/herglotz/.

Abstract

This report consolidates an experimental-mathematics study of the Herglotz function $F(x)$, building on Radchenko and Zagier’s paper. We do three things. First, we re-implement F , its companion $\mathcal{J}$, and the exact Theorem-3 dilogarithm sum independently (PARI/GP and **mpmath**) and re-verify *every* entry of the authors’ Tables 1 and 2 of quadratic-unit values to better than 10^{-80} ; two rows of Table 1 fail, and we identify them as genuine typographical errors in the printed paper, each fixed by a unique single edit. Second, we turn the paper’s two worked rational-argument examples into a family of general theorems: a closed form for $F(2/q)$ valid for all q (with proof), the pure-logarithm $F(1/q)$, the explicit $F(n/(n+1))$ family, a reciprocity law for $F(p/2)$, a derivative law for $F'(2/q)$, a clean collapse criterion ($F(p/q)$ reduces to rational-argument dilogarithms iff $\text{rad}(q) \mid 30$), and the determination that the paper’s pure-logarithm value $J(2/5)$ is special to $q \in \{3, 5\}$. Third, we connect the new constants to two adjacent corpora: the golden-argument dilogarithms of $F(2/5)$ to the golden polylogarithm ladders over $\mathbb{Q}(\sqrt{5})$, and the Clausen sum of the derivative law to the conductor- q Clausen lattice at roots of unity. Every displayed identity is verified numerically; a residual table is given in §5.

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1 The function and the exact dilogarithm sum

The Herglotz function may be taken in the integral form (the source paper's eq. (1))

$$J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t} dt, \quad J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}, \quad (1)$$

which relates the companion J to the Herglotz function F . For the rational-argument results the engine is Radchenko–Zagier's Theorem 3, an *exact, finite* evaluation obtained from the second integral representation of F via the distribution relations:

$$F\left(\frac{P}{Q}\right) - F(1) = \sum_{\alpha^P=1} \left(\sum_{\substack{\beta^Q=1 \\ \beta \neq 1}} f(\alpha, \beta) - \sum_{\substack{\beta^P=1 \\ \beta \neq 1}} f(\alpha, \beta) \right), \quad f(\alpha, \beta) = \text{Li}_2\left(\frac{\beta}{\beta-1}\right) - \text{Li}_2\left(\frac{\alpha-\beta}{1-\beta}\right). \quad (2)$$

Since (2) is a finite sum of dilogarithms it evaluates $F(P/Q) - F(1)$ to arbitrary precision with no quadrature error; throughout we use it as ground truth and cross-check against the defining integral (which is independent but only ~ 30 -digit accurate through `mpmath` quadrature).

2 Quadratic-unit values: Tables 1 & 2, and two errata

For real quadratic units the relevant object is the companion $\mathcal{J}(x) = J(x) - \frac{\log^2 2}{2} + \frac{\pi^2}{24}(x - \frac{1}{x})$, and the arithmetic atoms are $S_{p,q} = \log \varepsilon_p \log \varepsilon_q$ where ε_d is the fundamental unit of $\mathbb{Q}(\sqrt{d})$ for a fundamental discriminant $d > 1$ and $\varepsilon_1 := 2$. Tables 1 and 2 of the source paper express, for the 45 known one-class-per-genus cases, the quantities $4(\mathcal{J}(n + \sqrt{n^2 - 1}) - n\frac{\pi^2}{24})$ and $2(\mathcal{J}(n + \sqrt{n^2 + 1}) - n\frac{\pi^2}{24})$ as integer combinations of the $S_{p,q}$.

We re-implemented $\mathcal{J}$ and $S_{p,q}$ in PARI/GP (`intnum` integrals, `bnfinit` regulators) and evaluated both sides of all 45 rows at 200-digit precision. All 16 rows of Table 2 and 27 of the 29 rows of Table 1 agree to better than 10^{-80} . The two failures are $O(1)$:

$$n = 43: \text{LHS} - \text{RHS} = -13.0160\dots, \quad n = 61: \text{LHS} - \text{RHS} = -2.9011\dots$$

Each resolves to a single exact correction, confirmed against the high-DPI scan to be a typo in the printed paper rather than an OCR artifact.

Proposition 1 (Two corrections to Table 1).

- $n = 43$ ($n^2 - 1 = 2^3 \cdot 3 \cdot 7 \cdot 11$): the printed right-hand side overshoots by exactly $S_{33,56}$ ($(\text{LHS} - \text{RHS})/S_{33,56} = -1.000\dots$ to 110 digits); the spurious $+S_{33,56}$ term should be deleted.
- $n = 61$ ($n^2 - 1 = 2^3 \cdot 3 \cdot 5 \cdot 31$): the printed $S_{60,1032}$ should read $S_{60,248}$. The printed subscript $1032 = 2^3 \cdot 3 \cdot 43$ carries the prime 43, foreign to $n^2 - 1 = 3720$, whereas $60 \cdot 248 = 14880 = 4 \cdot 3720$ is the correct genus splitting.

A brute-force search over all single edits (delete a term, change a coefficient in $[-4, 4]$, or replace a subscript by any admissible fundamental-discriminant pair) returns these as the unique fixes. With both applied, all 45 rows verify to better than 10^{-80} .

A by-product of the same study maps where $\mathcal{J}$ ceases to be elementary: $\mathcal{J}(x)$ closes in $\{\zeta(2), \log 2 \log x, \log^2 x\}$ precisely for the even-trace family $x = n + \sqrt{n^2 \pm 1}$ (e.g. $\mathcal{J}(1 + \sqrt{2}) = \frac{1}{4}\zeta(2) + \frac{1}{2}\log 2 \log(1 + \sqrt{2})$), and *not* for a generic quadratic unit such as the golden ratio — the precise numerical content of the paper's remark that the individual values $F(x)$ have no general closed form.

3 Rational arguments: a family of theorems

The paper writes out only two rational-argument cases: $F(n)$ (pure logarithms, its eq. 23) and the lone fractional example $F(2/5)$ (its eq. 24). We now give the general laws.

3.1 Two reductions on the line $\operatorname{Re} = \frac{1}{2}$

Everything rests on one observation: for a unit $\beta = e^{i\theta}$,

$$\frac{\beta}{\beta-1} \text{ has real part } \frac{1}{2}, \quad \arg \frac{\beta}{\beta-1} = \frac{\theta}{2} - \frac{\pi}{2}, \quad \left| \frac{\beta}{\beta-1} \right| = \frac{1}{2 \sin(\theta/2)}. \quad (3)$$

By the reflection $\operatorname{Li}_2(z) + \operatorname{Li}_2(1-z) = \frac{\pi^2}{6} - \log z \log(1-z)$ with $1-z = \bar{z}$ on $\operatorname{Re} z = \frac{1}{2}$,

$$2 \operatorname{Re} \operatorname{Li}_2\left(\frac{1}{2} + iy\right) = \frac{\pi^2}{6} - \log^2 |z| - (\arg z)^2. \quad (4)$$

The second reduction handles the $\alpha = -1$ part of (2): $\frac{-1-\beta}{1-\beta} = -i \cot \frac{\theta}{2}$ is purely imaginary, and for real y

$$\operatorname{Re} \operatorname{Li}_2(iy) = \frac{1}{4} \operatorname{Li}_2(-y^2). \quad (5)$$

3.2 $F(1/q)$: pure logarithms

Theorem 2. For all $q \geq 2$, with $m = \lfloor (q-1)/2 \rfloor$,

$$F\left(\frac{1}{q}\right) - F(1) = -\frac{(q-1)(3q-2)}{24q} \pi^2 - \sum_{k=1}^m \log^2\left(2 \sin \frac{\pi k}{q}\right) - [q \text{ even}] \frac{1}{2} \log^2 2.$$

Proof. With $P = 1$ only $\alpha = 1$ occurs and the inner β^P -sum is empty, so (2) reads $F(1/q) - F(1) = \sum_{\beta^q=1, \beta \neq 1} (\operatorname{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6})$. Pairing β with $\bar{\beta}$ and applying (4) gives the π^2 coefficient $-\left[\frac{q-1}{12} + \sum_{k=1}^m \left(\frac{k}{q} - \frac{1}{2}\right)^2\right] = -\frac{(q-1)(3q-2)}{24q}$ and the log-sine sum. For even q the self-conjugate root $\beta = -1$ contributes $\operatorname{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2} \log^2 2$. $\square$

3.3 $F(2/q)$: the general theorem

Theorem 3. For every $q \geq 3$,

$$F\left(\frac{2}{q}\right) - F(1) = -\frac{(q-1)(q-2)}{12q} \pi^2 + \log^2 2 - 2 \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \log^2\left(2 \sin \frac{\pi k}{q}\right) - \frac{1}{2} \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \operatorname{Li}_2\left(-\cot^2 \frac{\pi k}{q}\right),$$

where for even q the two sums run to $k = \frac{q}{2} - 1$ and a further $-\log^2 2$ is added.

Proof (odd q). Here $\alpha \in \{1, -1\}$ and the inner β^P -sum is the single term $\beta = -1$. For the first part, $f(1, \beta) + f(-1, \beta) = 2 \operatorname{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6} - \operatorname{Li}_2(\frac{-1-\beta}{1-\beta})$. Summing over $\beta \neq 1$ and using (4) for the first pieces and (5) (with $\frac{-1-\beta}{1-\beta} = -i \cot \frac{\pi k}{q}$, pairing $k \leftrightarrow q-k$) for the last gives $2S_1 - (q-1)\frac{\pi^2}{6} - \frac{1}{2} \sum_k \operatorname{Li}_2(-\cot^2 \frac{\pi k}{q})$, where $S_1 = \sum_{k=1}^{(q-1)/2} [\frac{\pi^2}{6} - \log^2(2 \sin \frac{\pi k}{q}) - \pi^2(\frac{k}{q} - \frac{1}{2})^2]$. The boundary gives $-\sum_\alpha f(\alpha, -1) = +\log^2 2$ from $\operatorname{Li}_2(\frac{1}{2})$. Collecting, the $\frac{\pi^2}{6}$ terms cancel and the π^2 coefficient is $-2 \sum_{k=1}^{(q-1)/2} (\frac{k}{q} - \frac{1}{2})^2$, which equals $-\frac{(q-1)(q-2)}{12q}$ by an elementary identity (verified symbolically). The even- q case differs only in that $\beta = -1$ is now a genuine q -th root, contributing the extra $-\log^2 2$. $\square$

The coefficient is exactly twice that of eq. 23 for $F(q)$. The new ingredient $\operatorname{Li}_2(-\cot^2 \frac{\pi k}{q})$ is a dilogarithm at an algebraic argument of degree $\varphi(q)/2$: for $q = 5$ the two terms collapse (the real subfield is quadratic) to the single $\operatorname{Li}_2(4/5)$ of eq. 24, but for $q \geq 7$ they do not, which is why no simpler form than Theorem 3 exists there. A direct PSLQ search shows the sum $\sum_k \operatorname{Li}_2(-\cot^2 \frac{\pi k}{q})$ is itself irreducible over $\{\pi^2, \log^2 q, \text{Catalan}, \log\text{-sines}\}$.

3.4 The $F(n/(n+1))$ family and the $F(p/2)$ reciprocity

Theorem 4. For all $n \geq 2$, with $S(q) = \sum_{k=1}^{q-1} \log^2(2 \sin \frac{\pi k}{q})$,

$$F\left(\frac{n}{n+1}\right) - F(1) = -\frac{3n^2 + 3n + 2}{24n(n+1)}\pi^2 + \operatorname{Li}_2\left(\frac{n}{n+1}\right) + \frac{1}{2} \log^2 \frac{n+1}{n} - \frac{1}{2}S(n+1) + \frac{1}{2}S(n),$$

the single dilogarithm $\operatorname{Li}_2(n/(n+1))$ appearing with coefficient exactly $+1$ (the source paper's “bilinear-log” Example 2, made explicit). The π^2 coefficient equals $-\frac{1}{8} - \frac{1}{12n(n+1)}$.

Theorem 5 (Reciprocity). For all odd $p \geq 3$,

$$F\left(\frac{p}{2}\right) + F\left(\frac{2}{p}\right) - 2F(1) = -\frac{(p-2)^2}{12p}\pi^2 + \frac{1}{2} \log^2 \frac{p}{2}.$$

The $\operatorname{Li}_2(-\cot^2 \frac{\pi k}{p})$ terms of $F(p/2)$ are exactly minus those of $F(2/p)$ and cancel in the sum; combined with Theorem 3 this yields $F(p/2)$ in closed form.

3.5 The derivative law

Theorem 6 (Radchenko–Zagier Proposition 4, $P = 2$). For all $q \geq 3$, with $T(q) := \frac{2}{q}F'(2/q) - (1 + \log 2)$,

$$T(q) = \frac{q^2 - 4}{24q}\pi^2 + W(q), \quad W(q) = \sum_{k=1}^{q-1} \cot \frac{2\pi k}{q} \operatorname{Cl}_2\left(\frac{2\pi k}{q}\right),$$

with an extra $-\log 2$ for even q . Unlike $F(2/q)$ itself, this does not reduce to elementary form: $W(q)$ is a genuine Clausen combination.

3.6 The collapse criterion and the values $J(p/q)$

Proposition 7 (Collapse criterion). For coprime p/q , $F(p/q) - F(1)$ reduces to dilogarithms at rational arguments (together with π^2 and logarithms) if and only if $\operatorname{rad}(q) \mid 30$, i.e. every prime factor of q lies in $\{2, 3, 5\}$; otherwise it requires genuine Li_2 at the cyclotomic-ratio arguments $(1 - \zeta_p^a)/(1 - \zeta_q^b)$. Across all rational cases only two new dilogarithm constants ever occur: $\operatorname{Li}_2(1/3)$ (conductor 3) and the pair $\{\operatorname{Li}_2(1/5), \operatorname{Li}_2(2/5)\}$ (conductor 5).

Sample closed forms: $F(3/4) - F(1) = -\frac{19}{144}\pi^2 - \frac{3}{4}\log^2 2 - 2\log 2 \log 3 + \frac{7}{4}\log^2 3 + 2\operatorname{Li}_2(\frac{1}{3})$ and $F(4/5) - F(1) = \frac{3}{80}\pi^2 + \frac{11}{4}\log^2 2 - \frac{1}{2}\log^2 5 - \operatorname{Li}_2(\frac{1}{5}) - \log^2(2\sin \frac{\pi}{5}) - \log^2(2\sin \frac{2\pi}{5})$.

Proposition 8 (*J-values*). *With J as in (1), $J(2/q)$ is a pure-logarithm value if and only if $q \in \{3, 5\}$: $J(2/3) = -\frac{\pi^2}{144} + \frac{3}{4}\log^2 2$ and $J(2/5) = \frac{11\pi^2}{240} + \frac{3}{4}\log^2 2 - 2\log^2 \frac{\sqrt{5}+1}{2}$. So the paper's $J(2/5)$ — its only worked J example — is a special case, not a general law; the dilogarithms cancel only at $q = 3, 5$. The values $J(1/q)$ always retain the $\operatorname{Li}_2(-\cot^2)$ dilogarithms inherited from $F(2/q)$.*

4 Bridges to the golden ladders and the Clausen lattice

The new constants are not isolated; they touch two corpora we have already mapped.

The golden bridge ($q = 5$). The arguments $-\cot^2 \frac{\pi}{5} = -\frac{5+2\sqrt{5}}{5}$ and $-\cot^2 \frac{2\pi}{5} = -\frac{5-2\sqrt{5}}{5}$ lie in $\mathbb{Q}(\sqrt{5})$, the field of the golden polylogarithm ladders. Comparing Theorem 3 ($q = 5$) with eq. 24 gives the functional equation

$$\operatorname{Li}_2\left(\frac{4}{5}\right) + \operatorname{Li}_2\left(-\cot^2 \frac{\pi}{5}\right) + \operatorname{Li}_2\left(-\cot^2 \frac{2\pi}{5}\right) = 2\log^2 2 - 2\log 2 \log \frac{2}{5} - 6\left(\log^2(2\sin \frac{\pi}{5}) + \log^2(2\sin \frac{2\pi}{5})\right),$$

hence the golden-sum closed form (with $\varphi = \frac{1+\sqrt{5}}{2}$)

$$\operatorname{Li}_2\left(-\cot^2 \frac{\pi}{5}\right) + \operatorname{Li}_2\left(-\cot^2 \frac{2\pi}{5}\right) = \operatorname{Li}_2\left(\frac{1}{5}\right) - \frac{\pi^2}{6} - 3\log^2 \varphi + \frac{1}{4}\log^2 5.$$

Only the symmetric sum reduces: the antisymmetric part $X = \frac{1}{2}(\operatorname{Li}_2(-\cot^2 \frac{\pi}{5}) - \operatorname{Li}_2(-\cot^2 \frac{2\pi}{5}))$ admits no integer relation over $\{\pi^2, \log^2 \varphi, \log^2 5, \log^2 2, \operatorname{Li}_2(\frac{1}{5}), \operatorname{Li}_2(\frac{2}{5}), \log \text{ products}\}$, so it is a genuinely new $\mathbb{Q}(\sqrt{5})$ real-dilogarithm constant. (Consistently, X is not balanced as a Bloch-group element, so the vanishing of $K_3(\mathbb{Q}(\sqrt{5})) \otimes \mathbb{Q}$ does not force it to reduce.)

The Clausen bridge (the derivative law). The Clausen sum $W(q)$ of Theorem 6 is a $\mathbb{Q}(\zeta_q)^+$ -linear combination of the conductor- q Clausen values $\operatorname{Cl}_2(\frac{2\pi k}{q})$ — an element of the conductor- q Clausen lattice over the real cyclotomic field. The coefficients are rational over $\mathbb{Q}(\zeta_q)^+$ exactly when $\cot \frac{2\pi k}{q} \in \mathbb{Q}(\zeta_q)^+$, in which case $W(q)$ closes cleanly:

$$W(8) = 2\operatorname{Cl}_2\left(\frac{\pi}{4}\right) - 2\operatorname{Cl}_2\left(\frac{3\pi}{4}\right), \quad W(12) = \frac{22\sqrt{3}}{9}\left(\operatorname{Cl}_2\left(\frac{\pi}{6}\right) - \operatorname{Cl}_2\left(\frac{5\pi}{6}\right)\right).$$

For $q = 5, 7, 9, 11$ the cotangents lie in a strictly larger field, so no rational reduction exists, while lattice membership persists. Thus differentiating F at a rational moves from the real-dilogarithm world of $F(2/q)$ into the imaginary-dilogarithm (Clausen) lattice mapped at roots of unity.

5 Verification record

Every displayed identity was re-checked against the exact sum (2) (and, where applicable, the independent integral or a numerical derivative). The Theorem-1 derivation was additionally audited step-by-step (10^{-149}) and Theorem 2 confirmed to 10^{-159} by an independent multi-agent pass.

Result	test	max residual
Tables 1 & 2 (45 rows, 2 corrected)	PARI/GP, 200 digits	$< 10^{-80}$
Thm 2 $F(1/q)$	$q = 2..40$ vs (2)	1.0×10^{-99}
Thm 3 $F(2/q)$	$q = 3..40$ vs (2)	1.1×10^{-99}
Thm 4 $F(n/(n+1))$	$n = 2..16$	8.9×10^{-160}
Thm 5 $F(p/2)$ reciprocity	odd $p = 3..21$	4.1×10^{-159}
Thm 6 $F'(2/q)$	$q = 3..30$; vs numeric deriv.	1.1×10^{-129}
Prop 7 samples $F(3/4), F(4/5)$	PSLQ + canary	1.6×10^{-61}
Prop 8 $J(2/3), J(2/5)$	vs (2)	7.8×10^{-62}
§4 golden bridge / sum	vs (2)	1.6×10^{-61}
§4 $W(8), W(12)$	vs definition	1.6×10^{-60}

All PSLQ relations carried a Champernowne canary with coefficient 0 (no false positives). The reproducers are: `rational_harvest.py` (the exact sum and the proven formulas `F2q,F1q,Fn_over_np1,Fp2,dT2q`), `verify_full.gp` (the 45-row table check), `audit_thm1.py` (the step-by-step proof audit), and `bridges.py` (§4), all under `src/PolyLog/tools/scratch/herglotz/`. The verified statements are recorded in `src/PolyLog/identities/herglotz-rational-values.wl` and `herglotz-quadratic-units.wl`.

Summary

Starting from a paper that records the values of F at quadratic units (in two tables) and at rationals (through two worked examples), we have: independently re-verified the tables and caught two of their typos; promoted the two rational examples to a family of general theorems with proofs; and connected the resulting constants to the golden ladder field $\mathbb{Q}(\sqrt{5})$ and to the Clausen lattice at roots of unity. The Herglotz function is now a fully verified, well-connected part of the polylogarithm corpus.